Question Number etc. in left margin | Mark allocation in right margin

Q1 a i) Total calls per hour 280 calls/h → 4.66 calls/m
Average duration of call 3 m.
offered traffic = $4.66 \times 3 = 13.98$ Erlangs — 3

ii) $N = 21$ from traffic capacity Erlang chart — 4

iii) Loss probability = $B_c = 0.002$
Carried traffic = $13.98\,[1-0.002] = 13.95$ Erlangs — 3

Q1 b i) $\lambda = 173$ calls/h $\quad 1/\mu = 20$ s $\quad = E(S)$
$\lambda = 2.88333$ calls/m $\quad 1/\mu = 0.333$ m

$\rho = 0.96111$

$$E[Q_t] = \frac{\rho^2}{1-\rho} =$$ — 3

ii) $$E[W] = \frac{\rho}{1-\rho} E(S) =$$ — 3

iii) $$E[Q_t^2] = \sum i\,\rho(1-\rho)\rho^i = \rho \sum i^2 (1-\rho)\rho^i$$

$$= \rho\left(\rho\left(\frac{2}{(1-\rho)^2} - \frac{1}{1-\rho}\right)\right) = \rho^2\left(\frac{2}{(1-\rho)^2} - \frac{1}{1-\rho}\right)$$

$$Var[Q_t] = \rho^2\left(\frac{2}{(1-\rho)^2} - \frac{1}{1-\rho}\right) - \left(\frac{\rho^2}{1-\rho}\right)^2$$

$$= \frac{\rho^2}{(1-\rho)^2}\left(1 + \rho - \rho^2\right) =$$ — 4

Question Number etc. in left margin | Mark allocation in right margin

Q2 a) i)



Total arrival rate $\gamma = \lambda + s\gamma \rightarrow \gamma = \frac{\lambda}{1-s}$

If μ is the service rate per channel:

offered load per channel $= \frac{\gamma}{K\mu} = \frac{\lambda}{K\mu(1-s)}$

for a stable system: $\frac{\lambda}{K\mu(1-s)} < 1 \Rightarrow \frac{\lambda}{K\mu} < 1-s$

$\rho = \frac{\lambda}{K\mu}$ (offered traffic entering the system) $\Rightarrow \rho < 1-s \Rightarrow s < 1-\rho$ 4

ii) $K = 1$ a) without feedback $E(N_t) = \frac{\rho}{1-\rho}$ $\rho = \frac{\lambda}{\mu}$

a) with feedback $E(N_t) = \frac{\rho'}{1-\rho'}$ $\rho' = \frac{\gamma}{\mu} = \frac{\lambda}{\mu(1-s)}$

$= \frac{\rho}{(1-s-\rho)}$ 6

Q2 b) Link speed 64000 bits/s

message service rate $\mu = \frac{64000}{700}\ s^{-1}$

message arrival rate $\lambda = 80\ s^{-1}$ $\rho_H = \frac{\lambda}{\mu} = 0.875 = \rho_2$ 2

Acknowledgement service time $= \frac{2\times 32}{64000}\ s = 1\ ms$

$\rho_A = 80 \times 0.001 = 0.08 = \rho_1$

for a non-pre-emptive priority system: 2

mean message waiting time $= \frac{R}{(1-\rho_1)(1-\rho_1-\rho_2)} =$

$R = \frac{1}{2}\left[\lambda_1 E[S_1^2] + \lambda_2 E(S_2^2)\right]$

$\frac{1}{2}\left[\frac{\lambda_1}{\mu_1^2} + \frac{\lambda_2}{\mu_2^2}\right] = 4.83\ ms$

$= \frac{4.83}{0.92 \times 0.045}$

$= 0.1166\ s$ 6

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Q3 a) Mean value analysis of M/G/1 system

define R_i = residual service time = time until first departure seen by ith arrival

$$E[W_i] = E[R_i] + E[Q_i E[S]]$$
$$= E[R_i] + E[Q_i] E[S]$$

since Poisson arrivals see an unbiased sample of queue behaviour

$$E[W] = E[R] + E[Q] E[S] \quad \text{but also } E[Q] = \lambda E[W]$$

$$E[W] = \frac{E[R]}{1-\rho} \qquad \rho = \lambda E[S]$$

$$E[R_t] = \lim_{T\to\infty} \frac{1}{T} \int_0^T R_t \, dt$$

$$= \lim_{T\to\infty} \frac{1}{T} \sum_{i=1}^{M_T} \left(\frac{1}{2} S_i^2 \right) = \lim_{T\to\infty} \frac{1}{2} \left(\frac{M_T}{T} \right) \left[\frac{1}{M_T} \sum_{i=1}^{M_T} S_i^2 \right]$$

(M_T is the number of completed services in $[0,T]$)

$$E[R_t] = \frac{1}{2} \lambda E[S^2]$$

$$E[W] = \frac{\lambda E[S^2]}{2(1-\rho)}$$

4

for an M/D/1 system

S = constant = $h \Rightarrow E[S^2] = h^2$ $(\text{Var}(S) = 0)$

$$E[W] = \frac{\rho}{2(1-\rho)} h = \frac{\rho}{2(1-\rho)} E[S]$$

4

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Q3 b)

$$F_i(t+\Delta t, x) = [N-(i-1)]\lambda\Delta t\, F_{i-1}(t,x) + (i+1)\alpha\Delta t\, F_{i+1}(t,x) + \{1-[(N-i)\lambda+i\alpha]\Delta t\}F_i[t, x-\underbrace{(i-c)\alpha\Delta t}_{\Delta x}] + o(\Delta t)$$

$$-F_i(t,x) = -F_i(t,x)$$

$$F_i(t+\Delta t,x)-F_i(t,x) = h_1\Delta t\, F_{i-1}(t,x) + h_2\Delta t\, F_{i+1}(t,x) + h_3\Delta t\, F_i(t,x-\Delta x) + F_i(t,x-\Delta x) - F_i(t,x)$$

$$\frac{F_i(t+\Delta t,x)-F_i(t,x)}{\Delta t} = h_1 F_{i-1}(t,x) + h_2 F_{i+1}(t,x) + h_3 F_i(t,x-\Delta x) + \frac{F_i(t,x-\Delta x)-F_i(t,x)}{\Delta x}\frac{\Delta x}{\Delta t}$$

$\Delta t, \Delta x \to 0$

$$\frac{\partial F_i(t,x)}{\partial t} = h_1 F_{i-1}(t,x) + h_2 F_{i+1}(t,x) + h_3 F_i(t,x-\Delta t) - (i-c)\alpha\frac{\partial F_i(t,x)}{\partial x}$$

$h_1 = [N-(i-1)]\lambda$, $h_2 = (i+1)\alpha$, $h_3 = -[(N-i)\lambda+i\alpha]$ 6

ii) In statistical equilibrium: $\frac{\partial F_i(x,t)}{dt} = 0$ and $F_i(t,x) \to F_i(x)$

$$(i-c)\alpha\frac{dF_i(x)}{dx} = [N-(i-1)]\lambda F_{i-1}(x) - [(N-i)\lambda+i\alpha]F_i(x) + (i+1)\alpha F_{i+1}(x)$$

in matrix form:

$$\frac{dF(x)}{dx}D = F(x)M \Rightarrow \frac{dF(x)}{dx} = F(x)M'$$ 4

iii) This expression represents a set of first-order linear differential equations for which the solution is well-known to be a sum of exponentials 2

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Q4 a

i) ON-OFF source traffic source:

R_p = peak rate
$1/\beta$ = average burst length
$1/\alpha$ = average silence length
} probability source is ON $p = \frac{\alpha}{\alpha+\beta}$

Equivalent capacity $= C_L = mR_p + k\sigma R_p$

mR_p = mean $\Rightarrow m = Np$ for N ON-OFF sources

σR_p = standard deviation $\Rightarrow \sigma^2 = m(1-p)$ for N ON-OFF sources

$k = k(QoS)$

$$C = \frac{C_L}{R_p} = m + k\sigma$$
$$= Np + k\sqrt{Np(1-p)} \quad \text{and } k \sim P_L \text{ or } E$$

$$P_L = \sum_{i=J_0}^{N} \frac{(i-C)\pi_i}{m} \qquad E = \sum_{i=j}^{N} \pi_i$$

5

ii) For a large number of sources multiplexed, for $N \gg 1$, $p \ll 1$:

$$\pi_i = \binom{N}{i} p^i (1-p)^{N-i} \quad \text{(binomial)}$$

can be approximated quite closely by the normal distribution ($m = Np$, $\sigma^2 = Np(1-p)$)

$$E = \int_{J_0}^{\infty} \frac{e^{-\frac{(x-m)^2}{2\sigma^2}}}{\sqrt{2\pi\sigma^2}}\, dx$$

if $(C-m) > 3\sqrt{2}\sigma$

$$E = \frac{\sigma e^{-\frac{(C-m)^2}{2\sigma^2}}}{\sqrt{2\pi}\,(C-m)}$$

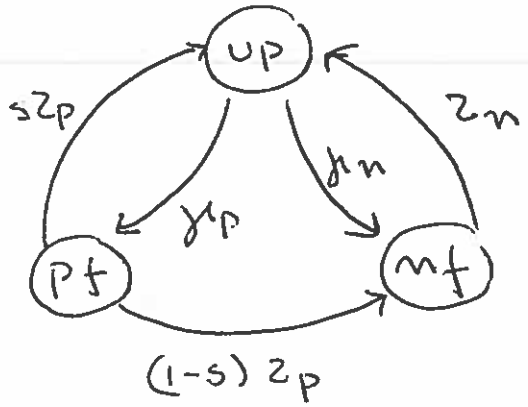
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Q4

a) i) $E = \{ up, \text{process failure}, \text{node failure} \}$ 3

ii)



4

iii)

	up	pf	nf
up	$-(\gamma_p+\gamma_n)$	γ_p	γ_n
pf	$s2_p$	-2_p	$(1-s)2_p$
nf	2_n	ϕ	-2_n

$= Q$

3